Quadrature Hybrid Coupler with Two Broadside Coupled Microstrip-Slot Lines

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Abstract — **This paper presents the design of a quadrature hybrid coupler, which is comprised of two broadside connected microstrip lines. The fixed coupling is obtained by a rectangular slot in the common ground plane of these lines. This simple coupler's structure enables coupling higher than 3dB in a wide frequency band. Such a coupler can be used in a lot of applications, especially in antenna beam-forming networks, measuring systems, power amplifiers and complex microwave integrated circuits (MIC's) with crosswise lines.**

Index Terms — ***wideband microwave components, quadrature hybrid coupler, microstrip-slot line***

I. Introduction

The coupler is a very simple and basic component of microwave integrated circuits (MIC's). Its implementation requires a very simple technology. Its attribute is also a low cost and weight. Moreover, such a structure can be implemented in a wide frequency band, up to the millimeter wave range [1, 2, 3]. The microstrip – slot quadrature hybrid coupler is a combination of two quarter wavelength sections of microstrip lines, which are broadside connected by grounds [4, 5]. Structurally, these grounds are the common ground plane for both microstrip lines. The coupling between two quarter wavelength microstrip lines is achieved by a slot in this common ground plane. This simple method of the implementation enables to obtain a very strong coupling, which is more than 3 dB. A small coupling slot enables the application of both small and a very small coupling.

In the paper it has been shown a simple way of designing and implementing the couplers. The value of the slot width has been achieved by a simple calculation and optimization of measuring method, which enables forming a coupler structure for any coupling.

Such a kind of the coupler can be used for the implementation of the power distribution in transmitters and measuring systems as well as for the construction of the Butler matrix as a linear and circular antenna beam forming networks [6]. The construction of double-layer enables to achieve the crosswise connecting lines inside integrated components without external connecting lines. The elimination of external connections with hold on specific phase shifts in a higher frequency band is a serious technological problem [7].

II. Structure of 3 dB Quatreture Hybrid Coupler

The structure of the quadrature hybrid microstrip-slot coupler has been shown in Fig. 1.

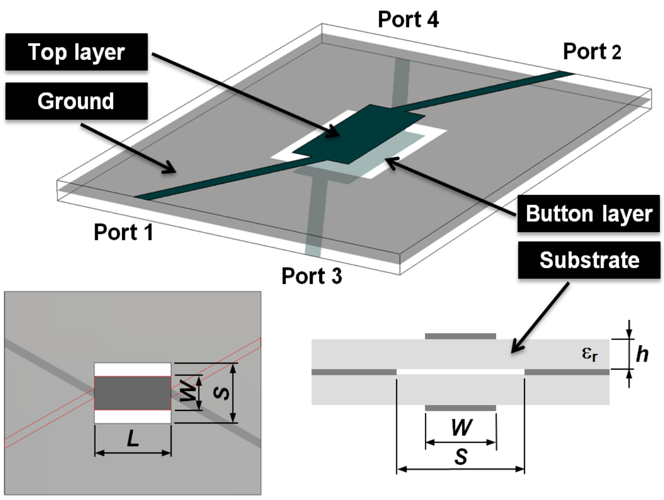

Fig. 1. The structure of microstrip – slot coupler

The coupler consists of two identical quarter wavelength microstrip lines with a common ground plane. The dimensions of top and button microstrip lines are W x L. The coupling between them is achieved by the coupling slot with dimensions S x L in the common ground. Therefore, it is the broadside coupling of two microstrip lines. Such approach enables strength coupling both microstrip lines by slots in ground plane. The value of this coupling is determined by the width of both microstrip lines W and the slot width S. The dimension of the microstrip width W and the slot length L are evaluated by a calculation the odd-mode characteristics impedance Z_{0o} for coupling lines.

III. The Design of 3 dB Coupler

The odd (Z_{0o}) and even (Z_{0e}) mode characteristics impedances for assuming characteristics impedance of the microstrip port of the coupler Z_0 and coupling coefficient k can be calculated from following equations :

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$$Z_0 = \sqrt{Z_{0o} Z_{0e}} \tag{1}$$

$$k = \frac{Z_{0e} - Z_{0o}}{Z_{0e} + Z_{0o}} \tag{2}$$

From the coupling coefficient k can be calculated the value of coupling factor C_{dB} as follows:

$$C_{dB} = 20 \log \left(\frac{Z_{0e} - Z_{0o}}{Z_{0e} + Z_{0o}} \right) \tag{3}$$

The odd-mode characteristic impedance Z_{0o} can be expressed by the characteristic impedance of a microstrip line, which has been shown as the upper part of cross-section of coupler in Fig. 2a.

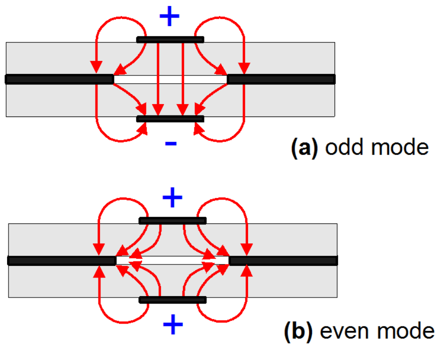


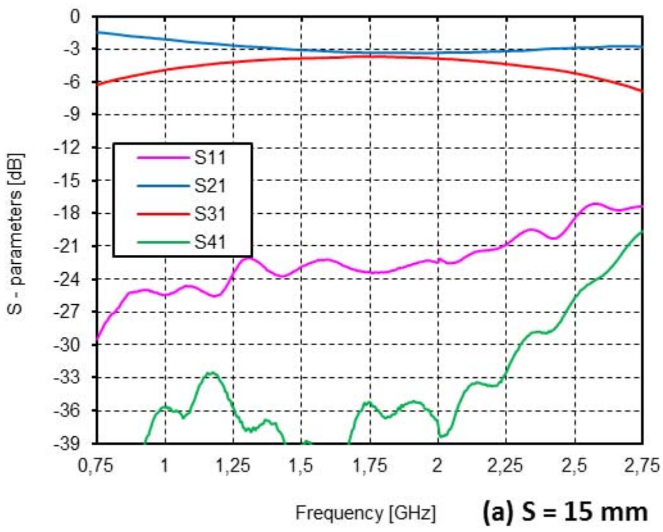
Fig. 2. The schematic expression of (a) odd – and (b) even – mode electric fields in the coupling region

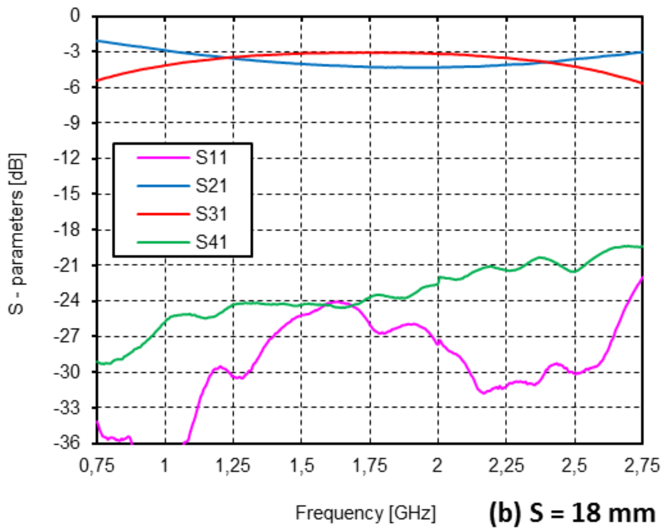
The synthesis method of such microwave lines is precisely described in references [8, 9, 10]. For the certain substrate thickness h and the characteristic impedance Z_{0o} of the microstrip line can be accurately estimated its width and length. Assuming that $Z_0 = 50$ ohms and the coupling factor C_{dB} is 3 dB, the value of $Z_{0o} = 20{,}7$ ohms. For such conditions it can be calculated the width of top and butter layer of the coupler. The length of coupling lines L also has been calculated for odd- mode and for quadrature hybrid coupler this dimension conforms to 90 degree electrical length.

The estimation of coupling slot width for the coupler is a more serious problem, because the analytical model of adequate lines is not determined. The proximate method to determine this slot width is the modified coplanar waveguide model. In our work this model has been used to a calculation of the initial value of slot width S. This accurate value has been achieved by the experimental approach.

IV. Measurement

Measurements of manufactured couplers with different coupling slot widths have been carried out in frequency range from 0.75 GHz to 2.75 GHz. Results of frequency characteristics measurement of the microstrip-slot quadrature hybrid coupler with three values of coupling slot width, have been shown in Fig. 3. The coupler has been constructed on FR4 substrate with thickness of h = 1,5mm and a dielectric constant of $\varepsilon_r = 4{,}6$.





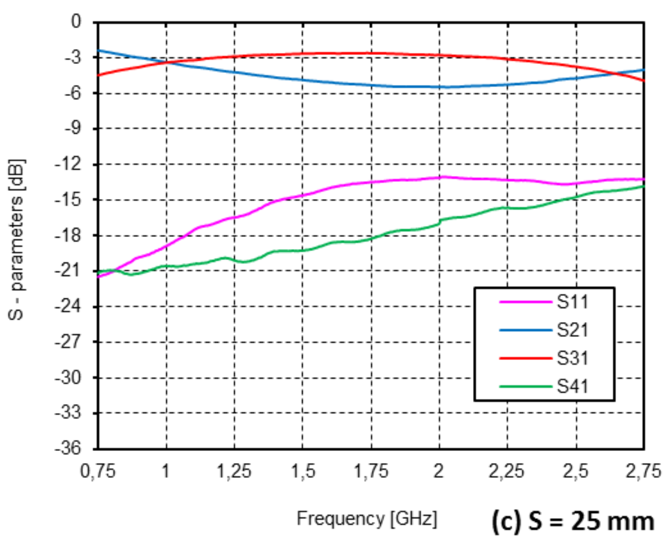


Fig. 3. Results of frequency characteristics measurement of microstrip-slot coupler with different slot widths: (a) S=15mm, (b) S=18mm, (c) S=25mm.

The coupling about - 3 dB has been achieved for the slot width of S = 18 mm. Simultaneously, the reflection coefficient has been the smallest and not overstep - 20 dB in almost whole frequency range. For the slot width of S = 25 mm impedance Z_{0e} has been greater. It has caused the increase of coupling over - 3 dB and the decrease of the inputs matching at the same time. For the slot width of S = 15 mm the coupling has decreased under - 3 dB and the inputs matching has decreased as well.

These results completely confirm the connection between the value of even mode characteristic impedance Z_{0e} and the slot width in the ground plane and its influence on the performance of the coupler. However, the optimal dimension of the slot width of S = 18 mm, which has been achieved by an experimental approach, is almost twice smaller than the value calculated for the coplanar waveguide.

The measured phase characteristic of the quadrature hybrid coupler with the slot width of S = 18 mm has been shown in Fig. 4.

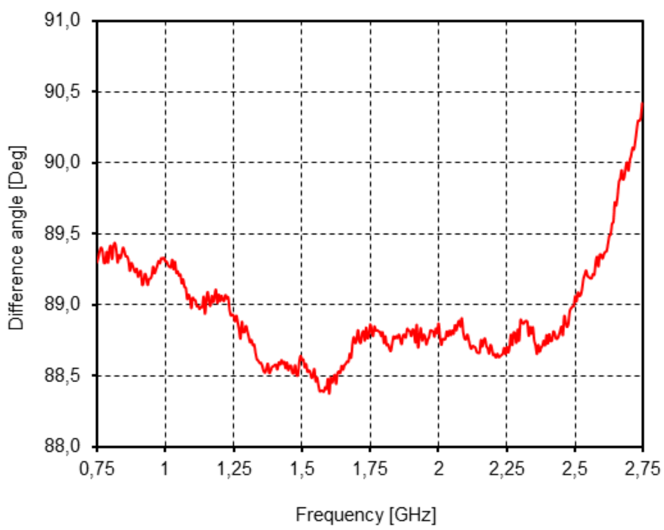


Fig. 4. The measured phase characteristic of the quadrature hybrid coupler with the slot width of S = 18 mm.

The phase difference between ports 2 and 3 of the coupler is 90° ± 1,5° in whole frequency range. At this stage of the research, it is difficult to determine if there are changes of coupler phase, or just measurement errors. This ambiguity is comparable with the accuracy of the measurement phase of the measurement set.

The implementation of the coupler with another coupling requires the change of the microstrip width, which can be calculated on the base analytical model of the microstrip line, but the slot width must be estimated by an experimental approach.

Research results of coupler, which have been shown in Fig. 3, suggest that the change of the slot width S affects the coupling without the change of the microstrip width W. It indicates that the choice of the microstrip width is not particularly critical.

V. Conclusion

In the paper the quadrature hybrid microstrip-slot coupler has been presented. On the base of manufactured circuits with different slot widths it has been carried out the analysis of the influence the geometry coupler on its performances. From this analysis, it is found out that the estimation of the slot width is critical issue for the coupler performance. Unfortunately, this parameter is also the most difficult to evaluate.

The simple structure of microstrip-slot coupler enables its implementation in a lot of microwave systems. Especially, in the circuits with certain phase ratios. It is the result of the elimination of external connections. Moreover, the double-side structure of this coupler enables its implementation in microwave integrated circuits with crosswise connections. It is very useful, especially for centimeter and millimeter wavelengths.

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